BUSINESS INTELLIGENCE REPORT

Health Insurance Charges Analysis

Dataset: 1,338 individual policy records

Generated: 26 March 2026

Analysis covers:

Smoking Status | Age Groups | Regional Breakdown | BMI Impact | Gender

Executive Summary

Total Customers	Avg. Annual Charge	Smoker Avg. Charge	Smoker Cost Ratio
1,338	**$13,270**	**$32,050**	**3.8x**
policy holders	across all customers	vs $8K non-smoker avg	more than non-smokers

• **Smoking is the #1 cost driver.** Smokers average $32,050 vs $8,434 for non-smokers - a 3.8x difference. With only 274 smokers (20% of customers), this group disproportionately drives claim costs.

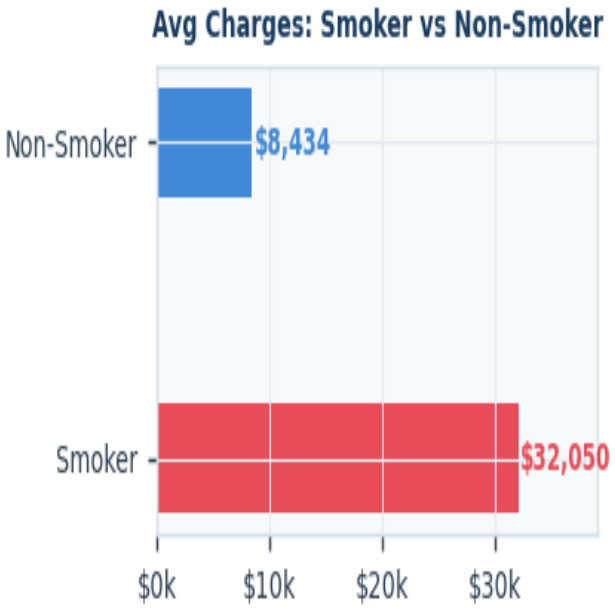
• **Age has a clear positive relationship with charges.** Seniors (55+) average ~$18.5K, nearly double the charges of customers under 25 at ~$9.0K.

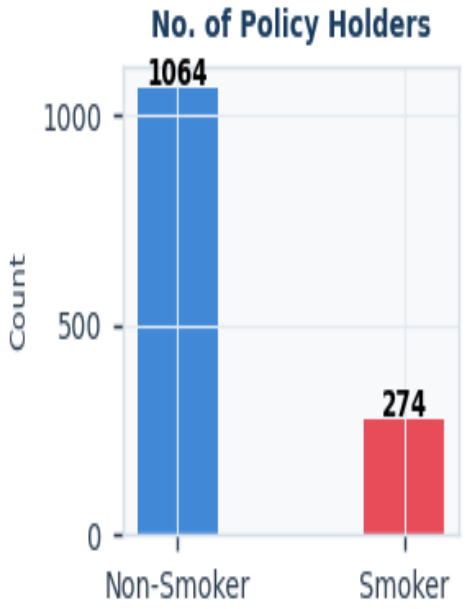
• **Regional variation is moderate.** The Southeast has the highest average charges (~$14.7K), while the Southwest averages around $12.3K.

• **BMI alone does not predict high charges.** Non-smokers maintain low charges across all BMI ranges. The high-cost cluster is almost entirely smokers, regardless of BMI.

• **Gender plays a minor role.** Males account for 50.5% of policies and average slightly higher charges than females, but the gap is small compared to smoking or age.

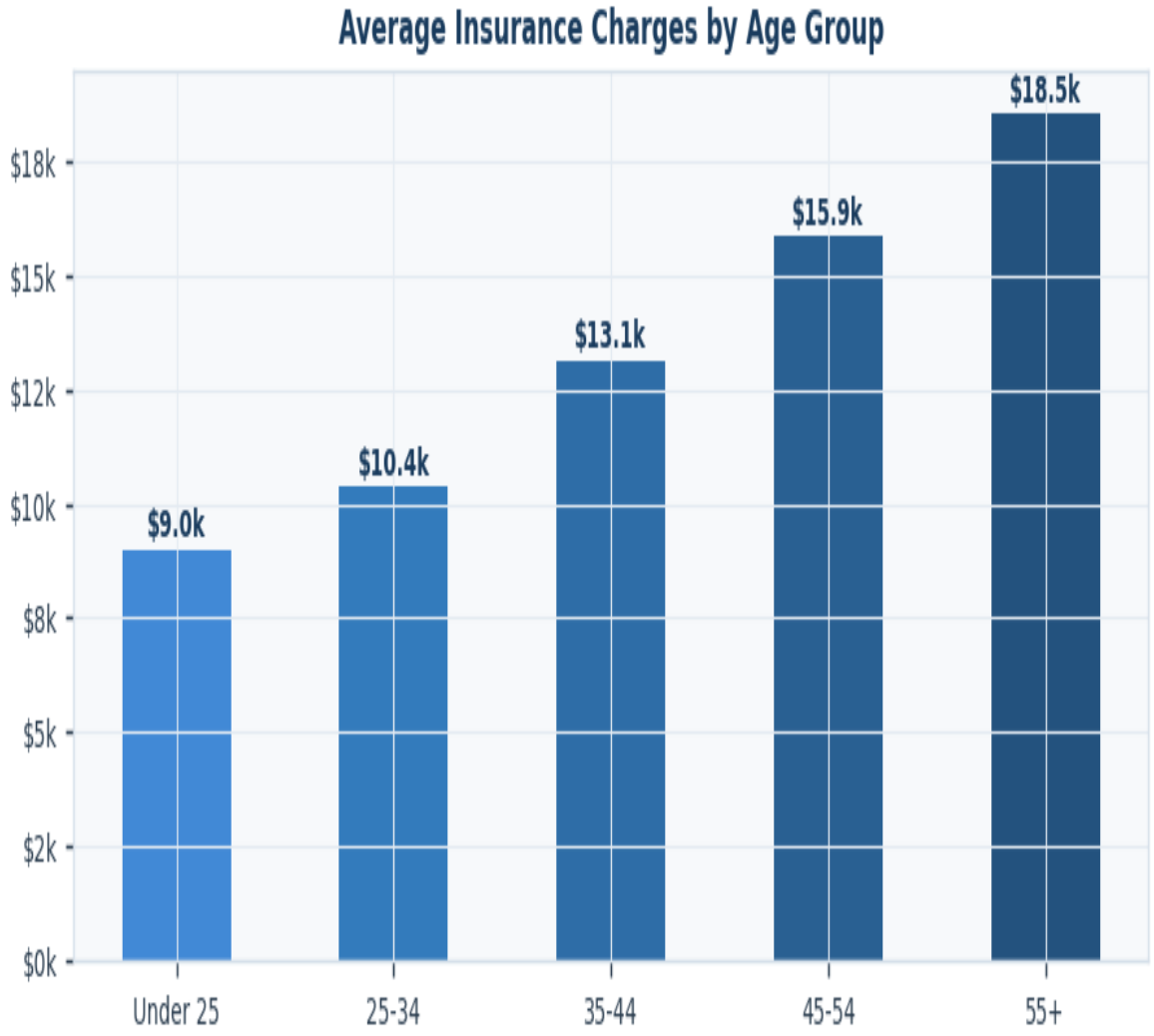
1. Smoker vs Non-Smoker Analysis





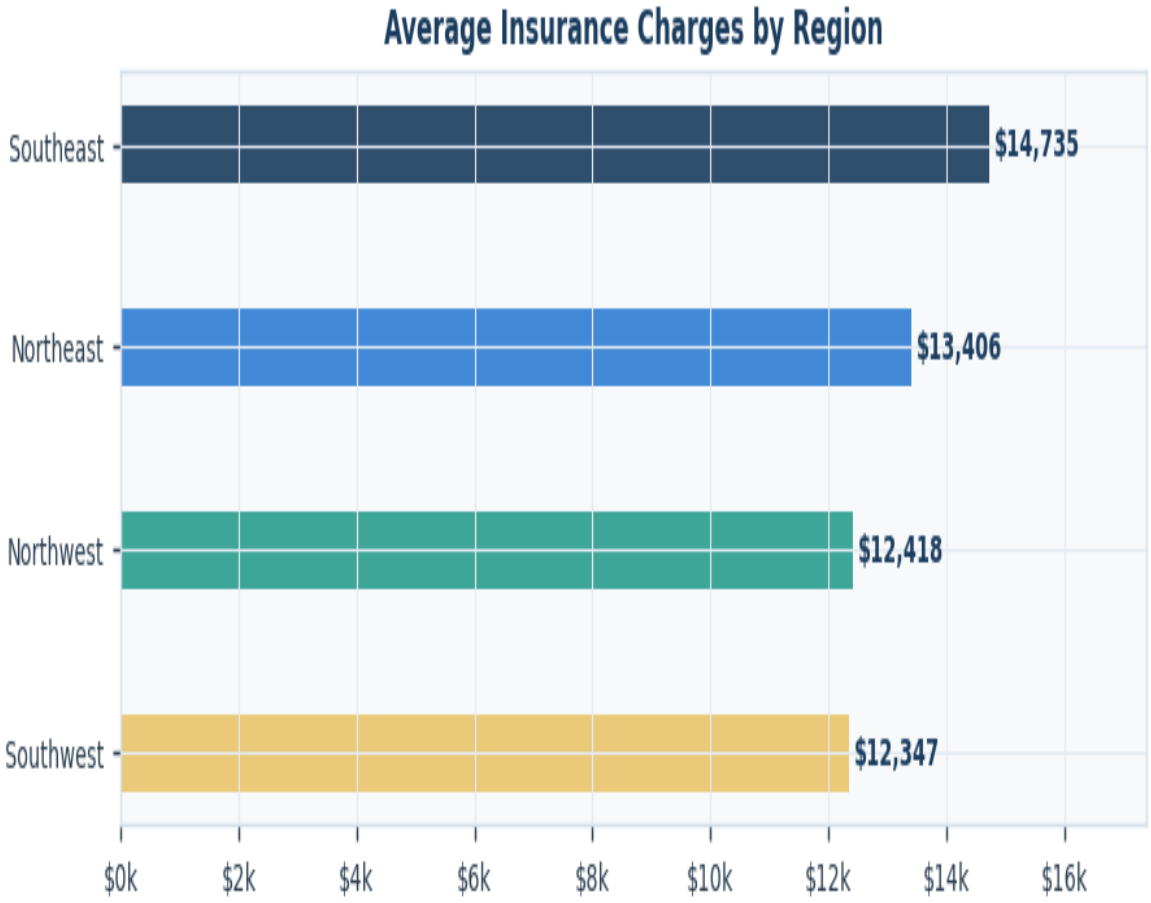
Status	Customers	% of Total	Avg Charge	Median Charge
Non-Smoker	1,064	79.5%	$8,434	$7,345
Smoker	274	20.5%	$32,050	$34,456

2. Charges by Age Group



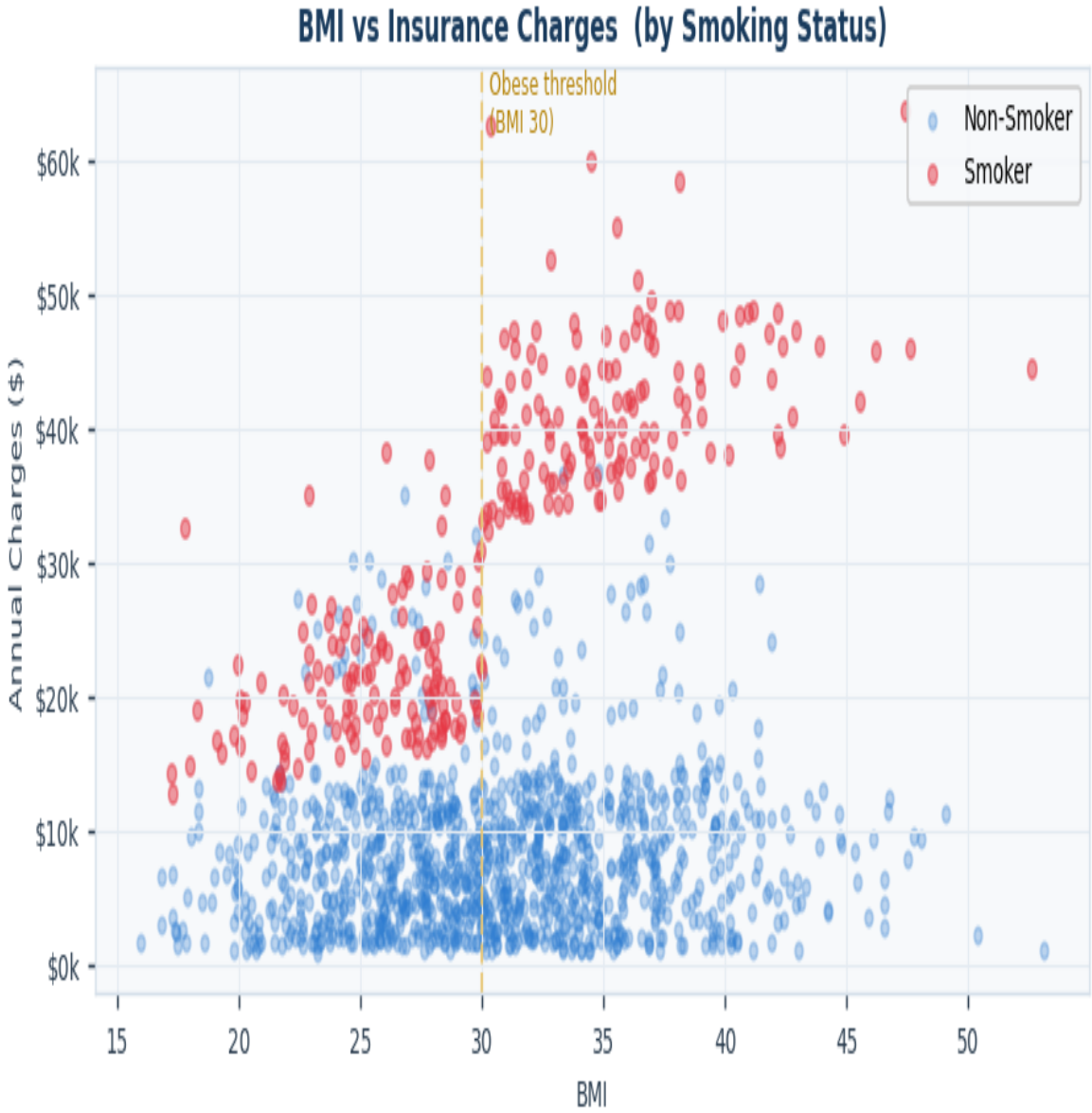
Age Group	Customers	Avg Charge	Median Charge	Index vs Overall
Under 25	278	$9,011	$2,457	68
25-34	271	$10,352	$4,718	78
35-44	260	$13,134	$7,206	99
45-54	287	$15,854	$10,115	119
55+	242	$18,513	$13,227	140

3. Regional Analysis



Region	Customers	Avg Charge	Median Charge	vs Overall Avg
Southeast	364	$14,735	$9,294	$+1,465
Northeast	324	$13,406	$10,058	$+136
Northwest	325	$12,418	$8,966	$-853
Southwest	325	$12,347	$8,799	$-923

4. BMI Impact on Insurance Charges

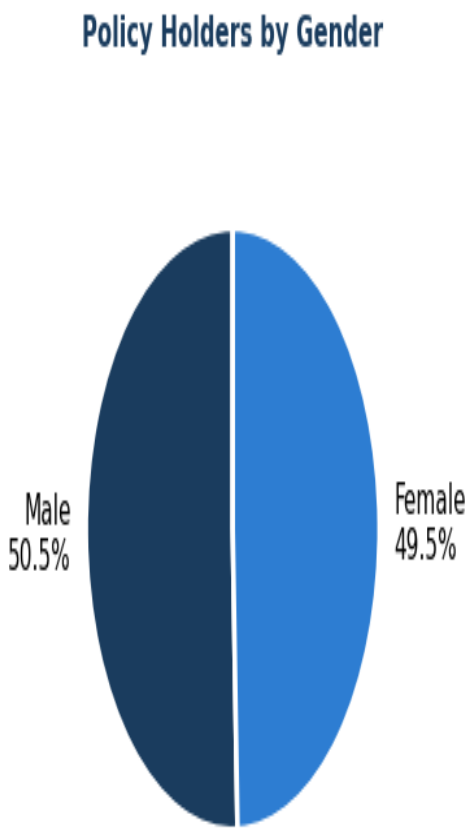


The scatter plot reveals two distinct populations. Non-smokers (blue) maintain relatively low charges across the full BMI range. Smokers (red) cluster at much higher charge levels regardless of BMI, demonstrating that smoking status dominates BMI as a cost predictor. The obese threshold (BMI 30) is shown as a dashed line for reference.

Average Charges by BMI Category and Smoking Status

BMI Category	Smoker Avg	Non-Smoker Avg	Overall Avg	Count
Underweight	$18,810	$5,485	$8,658	21
Normal	$19,942	$7,735	$10,435	226
Overweight	$22,491	$8,226	$10,998	386
Obese	$41,693	$8,853	$15,561	705

5. Gender Breakdown



Gender plays a minor role in insurance charges compared to smoking status, age, and region.

6. Top 10 Highest-Cost Policy Holders

All top 10 highest individual charges share a common profile: smokers, typically older age, often with higher BMI. This confirms smoking status as the primary risk indicator for extreme charges.

#	Age	Sex	BMI	Children	Smoker	Region	Annual Charge
1	54	Female	47.4	0	**Yes**	Southeast	$63,770
2	45	Male	30.4	0	**Yes**	Southeast	$62,593
3	52	Male	34.5	3	**Yes**	Northwest	$60,021
4	31	Female	38.1	1	**Yes**	Northeast	$58,571
5	33	Female	35.5	0	**Yes**	Northwest	$55,135
6	60	Male	32.8	0	**Yes**	Southwest	$52,591
7	28	Male	36.4	1	**Yes**	Southwest	$51,195
8	64	Male	37.0	2	**Yes**	Southeast	$49,578
9	59	Male	41.1	1	**Yes**	Southeast	$48,970
10	44	Female	38.1	0	**Yes**	Southeast	$48,885